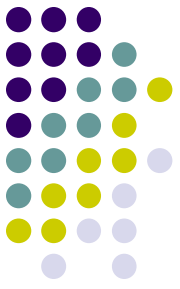


Chapter 8. Three-Level Models

Level – 1 Model (within classrooms)



$$Y_{ijk} = \pi_{0jk} + \pi_{1jk}a_{1ijk} + \pi_{2jk}a_{2ijk} + \dots + \pi_{Pjk}a_{Pijk} + e_{ijk},$$

Y_{ijk} is the achievement of child i in classroom j and school k

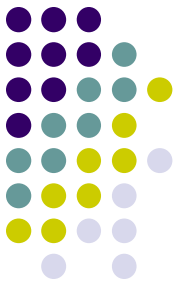
π_{0jk} is the intercept for classroom j in school k

a_{pijk} are $p = 1, \dots, P$ child characteristics

π_{pjk} are the corresponding level - 1 coefficients predicting the outcomes in classroom jk

e_{ijk} is a level – 1 random effect

$$e_{ijk} \sim N(0, \sigma^2)$$



Level – 2 Model: For each classroom effect,

$$\pi_{pjk} = \beta_{p0k} + \sum_{q=1}^{Q_p} \beta_{pqk} X_{qjk} + r_{pjk}$$

β_{p0k} is the intercept for school k

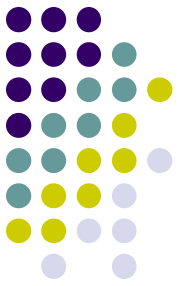
X_{qjk} is a classroom characteristic predicting effect π_{pjk} .
Each π_{pjk} may have a unique set of level-2 predictors

β_{pqk} is the corresponding level-2 coefficient.

r_{pjk} is a level-2 random effect

$$r_{pjk} \sim N(0, T_{\pi})$$

Level – 3 Model: Each level - 3 “outcome” may be predicted by some school-level characteristics



$$\beta_{pqk} = \gamma_{pq0} + \sum_{s=1}^{S_q} \gamma_{pqs} W_{sk} + u_{pqk},$$

γ_{pq0} is the intercept term in the school-level model for β_{pqk}

W_{sk} is a school characteristic predicting β_{pqk} . (Each β_{pq} may have a unique set of the level-3 predictors)

γ_{pqs} is the corresponding level-3 coefficient

u_{pqk} is a level-3 random effect, $u_{pqk} \sim N(\underset{\sim}{0}, \underset{\sim}{T}_{\beta})$